

Regional Quantitative Problem-Solving Circle

Problem Set 3: Number Theory & Modular Congruence (Weeks 5–6)

Lead Architect: Mohamed Jad Srifi

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Instructions: Do not rely on heuristic guessing. Every problem requires structural justification via Euclidean reductions, modular arithmetic properties, or Fermat’s Little Theorem. Ensure all parity arguments are formalized.

Problem 1 (Polynomial Divisibility & Euclidean Reduction — AIME Caliber). Find all positive integers n such that $(2n + 1) \mid (n^3 + 3n^2 + 7)$.

(Hint: Utilize synthetic division or polynomial Euclidean reduction to isolate an integer remainder that is independent of the variable n .)

Problem 2 (Order, Invariants & Fermat’s Little Theorem — Olympiad Track). Let p be an odd prime. Prove that any prime divisor q of the Mersenne number $2^p - 1$ must satisfy the congruence $q \equiv 1 \pmod{2p}$.

(Hint: Establish the multiplicative order of 2 (mod q), apply Fermat’s Little Theorem, and leverage the parity of primes.)

Problem 3 (Diophantine Integral Invariant — MathMaroc / Putnam Style). Determine all pairs of positive integers (x, y) satisfying the Diophantine equation $x^2 - 3y^2 = 17$.

(Hint: Analyze the algebraic constraint under a small modulo, such as modulo 3, and evaluate the set of possible quadratic residues.)